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GA-ConvE: An APT attack prediction method based on combination of graph attention network and 2D convolution

Yazhou Du, Weiwu Ren*, Wenjuan Li, Minyue Wang, Wanxiang Wang, Hewen Zhang, Mingqi Xia

School of Computer Science and Technology, Changchun University of Science and Technology, Changchun, 130022, China

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ABSTRACT

APT (Advanced Persistent Threat) attacks have become a significant challenge in the field of cybersecurity. Timely and accurate identification and prediction of APT attacks are crucial tasks. This paper proposes a new APT attack prediction method-GA-ConvE. By collecting APT threat intelligence and constructing an attack behavior knowledge graph, we classify and infer similar APT behaviors from a knowledge-based perspective, enabling effective prediction of APT attacks. To solve the problem in the classification module where some attack features are lost as the layers of the graph neural network increase, we propose a residual multi-layer graph attention network (RMultiGAT). Through this network leverages residual blocks, organizing and analyzing the data of multi-layer graph attention networks, we classify APT behaviors into different categories based on behavioral similarities. To tackle the challenges of large inference space, low prediction accuracy, and weak interpretability in the inference module, we introduce a joint prediction model combining graph attention networks and two-dimensional convolution (GA-ConvE). This model extracts behavioral features within each APT class and conducts targeted inferences for behaviors in each class, generating more precise embeddings, improving inference performance, thus realizing more accurate and interpretable predictions of APT attacks. Through Lots of experiments in real-world situations, the effectiveness of the GA-ConvE method in predicting APT attacks are validated. These research findings contribute to enhancing the real-time response capabilities against APT attacks.

1. Introduction

With the explosive growth of computer and network technologies, the types of network threats have proliferated, and cyberattacks are becoming increasingly complex, persistent, organized, weaponized and intelligent. It is challenging to identify the increasingly severe threats with a single identification method, especially in the case of Advanced Persistent Threat (APT) (Sharma et al., 2023b). According to article Akbarzadeh et al. (2024), the goal of APT attacks is to steal data, and their operations are often difficult to detect because these attacks can proceed slowly and continuously until their objectives are achieved. APT attacks are not one-time occurrences; they are continuous in nature (Sharma et al., 2023a). Numerous APT case studies have shown that once a network is infiltrated, the attackers will try to extract as much information as possible, and this behavior usually persists for months or even years (Li et al., 2023). Consequently, a single network breach can lead to continuous data leakage. Traditional threat defense systems (Abdulganiyu et al., 2023; Quincozes et al., 2021), such as Intrusion Detec-

tion Systems (IDS), Intrusion Detection and Prevention Systems (IDPS), and Advanced Security Appliances (ASA), respond to network attacks in a timely manner by monitoring network traffic, analyzing system activities, and identifying potential intrusion behaviors. However, APT attacks are characterized by their stealthiness, often employing novel techniques to hide their activities, which leads to a high false-positive rate in traditional threat prevention systems (Arefin et al., 2024). Therefore, there is an urgent need for timely, accurate, and reliable methods for detecting and predicting APT attacks.

APT attacks are typically characterized by long-term persistence. The multi-head attention mechanism of attention networks can focus on different parts of the input sequence, capturing these complex dependencies to better achieve organizational classification of APT attacks. However, as the number of attention layers increases, critical APT attack features may be partially lost, which are essential for accurate classification and directly impact the final classification results. To address this issue, residual connections (Szegedy et al., 2017) are introduced into deep neural networks, effectively allowing for the training of deeper

* Corresponding author.

E-mail addresses: Duyz129@outlook.com (Y. Du), renww@cust.edu.cn (W. Ren), 3289528441@qq.com (W. Li), 15143941806@163.com (M. Wang), umbrella1111W@163.com (W. Wang), CraneZhanghewen@outlook.com (H. Zhang), xmq1006@outlook.com (M. Xia).

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network structures and significantly improving the performance of deep models (Liu et al., 2025a,b). Models based on knowledge graph embeddings (Peng et al., 2023; Tang et al., 2024) can map entities and relationships into a low-dimensional vector space, achieving more compact representations and thereby reducing computational and storage overhead. These models can learn the semantic associations between entities and relationships, making them more flexible during inference and capable of capturing complex relationships within the knowledge graph. When facing data sparsity, knowledge graph embedding-based models not only handle sparse data more effectively but also support reasoning through the training of triples (Rosso et al., 2020), making them suitable for knowledge graph completion and APT attack prediction tasks, while providing higher interpretability during the inference process. This interpretability helps in understanding attacker behavior and deeper intelligence reasoning. Improving the efficiency of APT attack detection currently faces several challenges and difficulties, including the lack of publicly available APT attack data (Stojanovic et al., 2020), data imbalance (Benabderrahmane et al., 2024), and the use of standard encoding protocols (Xuan & Dao, 2021). Moreover, relying solely on security databases is insufficient for the organizational classification of APT attacks, as the software security entities and their relationships within APT attacks are often concealed within document contents and hyperlinks. However, by mining the relationships between disclosed APT threat intelligence, such as Common Vulnerabilities and Exposures (CVE), Common Weakness Enumeration (CWE), Common Attack Pattern Enumeration and Classification (CAPEC), and Indicators of Compromise (IOC). CVE provides a unique identifier and description for known vulnerabilities, allowing security teams to quickly locate and remediate those frequently exploited by APTs (Aggarwal, 2023). CWE summarizes the common weakness types underlying vulnerabilities, thereby assisting developers and auditors in reducing exploitable security flaws at the source (Singh et al., 2024). CAPEC, from the attacker's perspective, categorizes common attack patterns, enabling defenders to simulate, detect, and attribute the attack techniques employed by APT groups (Riera et al., 2022). IOC represents a critical component of threat intelligence, as it facilitates the rapid identification of malicious activities within technical infrastructures. It enables the precise description of the techniques employed by threat actors (such as tools or artifacts) as well as the Tactics, Techniques, and Procedures (TTPs) they develop (Villalón-Huerta et al., 2022). Hash values are an important part of IOC, commonly used to uniquely identify files, malware, or other data. For example, the hash value 92480c1a771d99dbef00436e74c9a927 represents `infsvc.exe`, which can use the CVE-2017-8570 vulnerability to bundle a trojan onto an OFFICE file. Similarly, `dbcfe5f5b568537450e9fc7b686adffd` corresponds to `taskhost.exe`, whose primary function is to gather information and then write it to the sectors of removable disks; `03bd34a9ba4890f37ac8fed78feac199` refers to `bindsvc.exe`, which can infect other executable files on the host and enable worm-like behavior (Forum, 2025). Domains and URLs in IOC indicators are also closely tied to attack techniques. For instance, a report (Column, 2025) shows that after `msheader2.exe` runs, it initializes by transferring files, executing shell commands, and establishes a connection with `45.11[.]19.170:80`. This report also demonstrates that a malicious sample exploits the CVE-2018-0798 vulnerability to execute shellcode, create scheduled tasks, and download an MSI file from <http://axopcustomersvc.com/gt.php> for execution. By associating and modeling multi-source security intelligence such as CVE, CWE, CAPEC, and IOC in the form of graph structures, it is possible to overcome the limitations of fragmented information within individual databases and achieve a multi-level unified representation of CVE vulnerabilities, CWE weaknesses, CAPEC attack patterns, and IOCs, thereby providing a complete depiction of the APT attack chain. Graph-based methods not only reveal potential relationships among different security entities but also enable the discovery of structural similarities in APT attack paths through graph neural networks. Therefore, in this study, various security entities (Abbas-Escribano & De-

bar, 2023)(CVE, CWE, CAPEC, and IOCs) are interconnected in a graph structure, which allows for a more comprehensive representation of the overall landscape of APT attacks and facilitates more accurate prediction of APT activities.

Due to the persistent nature of APT attacks, the multi-head attention mechanism is particularly suited to focusing on different parts of the input sequence, capturing complex dependencies. The application of residual connections helps retain critical attack features, thereby facilitating the organizational classification of APT attacks. Additionally, ConvE has demonstrated outstanding performance in predicting entity relationships in knowledge graphs, effectively reasoning about APT attackers' intelligence and enhancing the interpretability of results. Thus, this paper first extracts the relationships between entities from threat intelligence and generates triples according to their corresponding relationships, constructing a cybersecurity knowledge graph (Sikos, 2023). Subsequently, a multi-layer graph attention network model with residual connections is employed to perform organizational classification of APT attacks. Following this, a 2D convolutional link prediction model is used to conduct targeted reasoning and prediction of the classified APT attacks. By adopting this classify-then-predict strategy, the reasoning space is narrowed, inference speed is improved, and irrelevant knowledge is effectively filtered out, enhancing the accuracy of inference. This approach not only enables the organizational classification of APT attacks but also facilitates deeper-level attack predictions.

The main contributions of this paper are as follows:

- (1) Introducing residual blocks into multi-layer Graph Attention Networks, proposing RMultiGAT, and innovatively applying it to APT organization classification.
- (2) Combining multi-layer attention networks with residual connections and a two-dimensional convolutional link prediction model, we propose an APT attack prediction method based on the combination of a Graph Attention Network and two-dimensional convolution (GA-ConvE).
- (3) By correlating APT organizations, CVE, CWE, CAPEC, and IOCs data, a dataset for APT attack detection and reasoning is constructed. Using GA-ConvE, we conduct the first APT attack prediction experiment, achieving both APT organization classification and deeper intelligence reasoning.

2. Related work

In the field of cybersecurity, Advanced Persistent Threat (APT) have emerged as a sophisticated type of attack that is difficult to detect and defend against due to their stealthiness and persistence. APT attacks typically exhibit complex characteristics, involving multiple stages and techniques. Traditional detection methods based on feature matching and rule-based systems have proven to be insufficient in dealing with these complex attacks. As a result, researchers have begun exploring more intelligent detection methods, particularly techniques using neural networks (Omer et al., 2023), graph models (Dong et al., 2023), and graph neural networks (Bilot et al., 2023), to improve the accuracy and efficiency of APT attack detection and prediction.

In neural network-based models, Zhou and Huang (2024) proposes a method combining ensemble learning and convolutional neural networks (CNN), which was applied to attack detection in recommendation systems for the first time. This method integrates Bagging with CNN, enabling the automatic learning of complex patterns in user behavior data and improving the model's generalization capability. Yin et al. (2024) introduces the BTCN method, a detection approach based on bidirectional temporal convolutional networks. This method captures the dependencies before and after log events through forward and backward temporal convolutional layers, and by combining labeled abnormal data with unlabeled normal data in a semi-supervised training manner, it effectively utilizes sparse data, significantly enhancing the ability to identify abnormal patterns. A detection method based on Intelligent

Table 1
Comparison of GNN-based methods.

Approaches	Duan et al. (2023)	Tang et al. (2023)	Ren et al. (2023)	Zhao et al. (2024)	GA-ConvE
Attack Detection	✓	✓	✓	✓	✓
Attack Classification	✓	✓	✗	✗	✓
Dependency Modeling	✓	✓	✓	✓	✓
Reasoning Explainability	✗	✗	✗	✗	✓

Cognitive Computing is proposed in article [Xuan et al. \(2023\)](#), marking the first time that intelligent cognitive computing has been applied to software vulnerability detection. By integrating techniques such as machine learning, natural language processing, pattern recognition, and big data analysis, Intelligent Cognitive Computing improves both detection accuracy and efficiency. [Li et al. \(2024a\)](#) introduces the GMADV framework, consisting of two components: a malware variant generator and an adversarial training classifier. The generator enhances the diversity and quantity of data, while the classifier maintains a high detection rate even when confronted with new or unknown variants, greatly improving the robustness and generalization of the detection method. [Lu and Wang \(2023\)](#) proposes the DOMR method, which leverages the powerful feature learning capabilities of deep learning models by integrating multiple models into one framework. A novel training strategy is introduced, allowing the model to effectively learn how to recognize known categories while remaining sensitive to unknown categories, thus improving the overall detection capability of the system. From the above content, it can be observed that these neural network-based methods excel in feature extraction and handling complex data patterns. By employing techniques such as convolution, temporal convolution, multi-model integration, and adversarial training, they enhance the model's generalization and robustness. However, they fall short in capturing structured relationships and handling complex topological structures.

In graph model-based methods, the T-trace framework appears in article [Li et al. \(2024b\)](#), which constructs an APT attack provenance graph by analyzing the correlations in multi-source system logs. After correlation analysis, T-trace can identify the various stages of an APT attack and reconstruct the attack path. This framework provides a novel approach for APT attack detection and path reconstruction. The CoGraph method is introduced by the author of article [Li and Chen \(2024\)](#), which combines provenance graphs with process context. By analyzing provenance graphs and employing a multi-level detection strategy, this method offers a new approach for APT attack detection in CPS environments. From the above, it is evident that graph model-based methods have significant advantages in handling complex dependency relationships and reconstructing attack paths. However, they lack the capability to handle unstructured data, automated feature learning, and other aspects.

Compared to neural networks and graph models, graph neural networks (GNNs) can effectively capture complex dependencies and topological information in graph structures through automated feature learning and message passing mechanisms, automatically extracting multi-level relationships between nodes, making them suitable for processing large-scale complex networks ([Wu et al., 2021](#)). In recent years, methods based on GNNs have been developed. [Duan et al. \(2023\)](#) comes up with the DLGNN detection method, which combines dynamic line graph neural networks with semi-supervised learning. This method dynamically updates the line graph structure and improves model generalization. Furthermore, DLGNN analyzes graph structure information in network traffic, enabling it to identify complex attack patterns and behaviors that are difficult to capture with traditional methods, thus enhancing detection comprehensiveness and accuracy. [Tang et al. \(2023\)](#) recommends a vulnerability detection method based on GGNN. By iteratively propagating and updating graph structure data, GGNN updates each node's representation in every iteration based on the current state of the node and its neighboring nodes. The introduction of attention mechanisms effectively handles complex dependencies and contextual information in code, significantly improving the accuracy and efficiency

of vulnerability detection. [Ren et al. \(2023\)](#) puts forward a detection method based on graph convolutional networks (GCNs). This method constructs APT attack graphs by defining entities and relationships and uses GCN models to identify complex relationships in APT attacks that are difficult to detect with traditional models, making it suitable for network environments and security scenarios of varying scales. Article [Zhao et al. \(2024\)](#) employs causal inference and an attention-based encoder-decoder model to generate attack scenario graphs, followed by using GNNs to extract features from the graph for attack prediction. A detailed comparison between the four GNN-based methods and the method proposed in this paper is presented in [Table 1](#). From the above research, it can be concluded that although current methods can flexibly learn graph structures and integrate features of nodes and edges, they struggle to simultaneously meet the requirements of both attack detection and classification, as well as the demand for result interpretability.

3. Overall framework

A model for predicting APT attacks, which combines graph attention network and 2D convolutional linkage, is proposed in this paper. The process of how the model handles data and derives results is explained.

In the classification phase, this paper innovatively proposes RMulti-GAT (Residual Multi-Layer Graph Attention Network), which combines the multi-layer graph attention mechanism and residual connections to enhance the model's learning capability and stability when dealing with graph data. It is applied to the attack classification of APT organizations. This approach allows for the flexible capture of node relationships when processing graph-structured data and is suitable for various real-world scenarios such as image classification and text classification.

In the reasoning phase, ConvE (Convolutional 2D Knowledge Graph Embeddings) is a model used for knowledge graph embeddings based on convolutional neural networks, primarily designed to learn compact representations of entities and relationships in the graph. By introducing 2D convolution operations, this model effectively captures the semantic information between entities and relationships. ConvE has demonstrated strong performance in handling entities and relationships within knowledge graphs, especially in identifying patterns with spatial locality, and it has been widely applied to tasks such as knowledge graph completion. In this paper, the ConvE model is used to map relationships between constructed APT attack entities into low-dimensional vector representations. Through the embedding layer, security entities and relationships are mapped into real-valued vectors. The 2D convolution layer captures local patterns, the fully connected layer integrates features, and finally, the sigmoid function is used to predict APT attacks.

The overall framework of this paper consists of three parts, as shown in [Fig. 1](#). The first part, data processing, focuses on entity extraction and knowledge graph construction. It involves collecting vulnerabilities used during APT attacks and the IOCs tags generated by APT attacks from various APT reports. These vulnerabilities are then linked to relevant weaknesses and attack methods. Triples are constructed based on the relationships between entities, and a knowledge graph is built accordingly. The second part, APT attack classification, involves constructing a feature matrix and the corresponding adjacency matrix based on the relationships between entities. RMultiGAT is then used to classify these relationships. The third part, APT attack prediction, takes the classified APT attack data and inputs them into the ConvE model to predict APT attacks. The prediction process involves step-by-step reasoning, starting

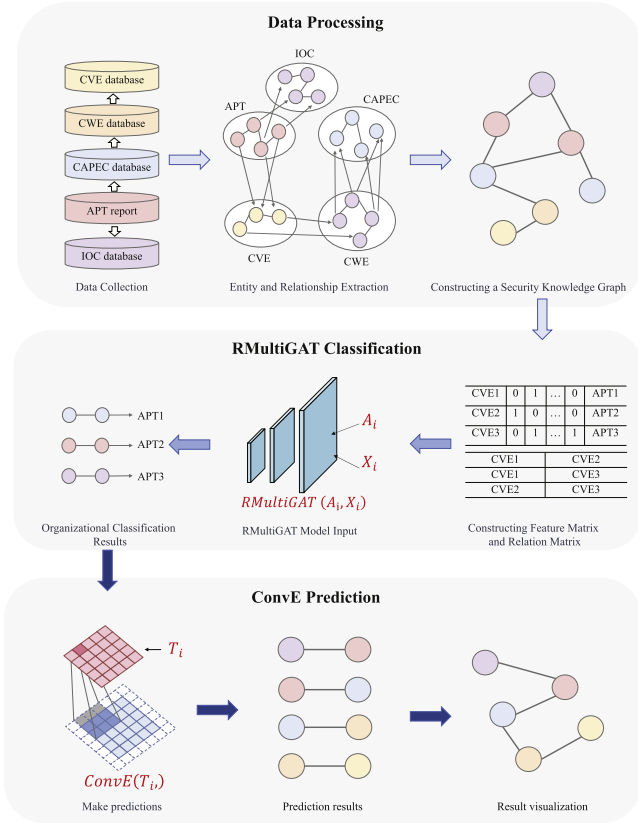


Fig. 1. Overall framework diagram.

from the IOCs to identify the source APT organization, then using the APT organization to infer potential CVE vulnerabilities, followed by predicting the CWE weaknesses, and finally forecasting the CAPEC attack methods. This approach presents a comprehensive view of the predicted APT attack.

4. Knowledge graph construction

The knowledge graph constructed in this paper primarily includes APT organizations, IOCs tags, CVE vulnerabilities, CWE weaknesses, and CAPEC attack methods. Based on publicly available APT threat intelligence from the web, relationships between APT organizations and CVE vulnerabilities, as well as the IOCs tags generated by the attacks, are extracted. Corresponding relationships between CVE vulnerabilities, CWE weaknesses, and CAPEC attack methods are then scraped from the MITRE website. The knowledge graph is constructed based on these corresponding relationships, and it is introduced in the following two subsections.

4.1. Entity extraction

Due to the covert nature of APT (Advanced Persistent Threat) attacks, obtaining relevant data is particularly challenging. In constructing the aforementioned cybersecurity knowledge graph for APT attacks, this paper collected publicly available reports on 12 relatively active APT groups. These groups encompass a wide range of attack methods and affected industries, with several of them actively launching attacks in recent times. Using regular expressions, CVE vulnerability entities and IOC (Indicator of Compromise) tags were extracted from the typical APT reports. CWE entities and CAPEC entities were extracted by mapping the CVE entities to data scraped from the MITRE website. As shown in Fig. 2, an example set of real APT attack entities was extracted using APT32 as a reference. By analyzing publicly available attack reports related to

APT32, CVE and IOC entities were identified. Subsequently, corresponding CWE and CAPEC entities were extracted based on their relationships. Each type of entity exists independently within its respective category.

4.2. Triple construction

By identifying the CVE vulnerabilities and IOCs tags used in APT attacks from APT threat reports, the relationships between APT attacks, IOCs tags, CVE vulnerabilities, CWE weaknesses, and CAPEC attack methods are associated. These relationships are then stored in the format of $\langle \text{Entity}, \text{Relationship}, \text{Entity} \rangle$. The following outlines the relationships between the identified entities and the resulting triples.

IOCs are traces left by APT attacks and are crucial for detecting APT attacks. Therefore, the relationship between the two is defined as $\langle \text{IOCs}, \text{Associated}, \text{APT} \rangle$. An APT attack may exploit CVE vulnerabilities as one of its means of attack, so the relationship between them is defined as $\langle \text{APT}, \text{Exploits}, \text{CVE} \rangle$. Typically, a single CVE vulnerability may involve multiple CWE weaknesses, so the relationship between them is defined as $\langle \text{CVE}, \text{Has}, \text{CWE} \rangle$. A CWE weakness may be associated with multiple CAPEC attack methods, so the relationship between them is defined as $\langle \text{CWE}, \text{Related}, \text{CAPEC} \rangle$. The relationships between entities such as CVE, CWE, and CAPEC are primarily derived from research and databases by security organizations such as MITRE. By centering on CWE, we can extract the associated CVE and CAPEC. Additionally, APT attacks often generate a series of IOCs and exploit CVE vulnerabilities. Therefore, the definition of relationships between these entities is based on real-world experience. From APT reports, we can identify the relationships between APT, CVE, and IOCs. By linking the relationships between these entities, we can construct a complete chain. For example, Fig. 3 shows a partial knowledge graph constructed from an APT32 attack dataset, where $\langle 641b7ab0d42d283f5b34de84cdcdcd6, \text{Associated}, \text{APT32} \rangle$, $\langle \text{APT32}, \text{Exploits}, \text{CVE-2021-22986} \rangle$, $\langle \text{CVE-2021-22986}, \text{Has}, \text{CWE-918} \rangle$, and $\langle \text{CWE-918}, \text{Related}, \text{CAPEC-664} \rangle$ are instances of constructed triples.

5. GA-ConvE prediction method

This paper proposes the GA-ConvE prediction method, which utilizes a multi-layer Graph Attention Network with residual connections to learn from the constructed security entities, thereby enabling the classification of APT attacks. After classification, the security entities from the APT attack data are formed into triples, and a 2D convolutional link prediction model is used to train and predict these triples.

The GA-ConvE method proposed in this paper leverages RMultiGAT to consider complex relational information during APT classification. By using graph structures, hidden attack patterns and structures are mined, achieving multi-level feature learning for more accurate APT attack classification. The ConvE model employs convolutional neural networks to capture entity and relationship information. Since convolutional neural networks can progressively combine and extract features through stacked convolutional layers, this mechanism allows the model to build more complex high-level semantic features from simple low-level features, thereby enhancing the understanding of semantic relationships and providing better interpretability. This makes it well-suited for handling complex APT attack relationships. The approach of first classifying APT attacks and then predicting them helps to reveal the full scope of APT attacks in a targeted manner, improving understanding of their behavior, accelerating inference, and increasing accuracy while reducing the potential impact of threats on organizations. Therefore, this paper proposes the RMultiGAT model for classifying APT attacks, followed by the ConvE model for predicting the classified attacks.

5.1. Node feature extraction

The data used in this paper is primarily constructed from publicly available online data and the relationships of real APT attacks. After

On the Linux platform, we have identified three types of trojans with architectures x86, Arm, and Mips, respectively. Through sample analysis, we have determined that these trojans are derived from a common set of source code through cross-compilation. Their communication protocols are entirely identical, and we have designated them as the "caja" trojan.

IOC Entity Instances

Md5	Architecture
641b7ab0d42d283f5b34de84cdcdcd6	Arm
a2d70e7ab7dccc5efcc32b5bbfdecad9	Mips
a54330bc0fdc9c9585f6024dde340177	X86

CVE Entity Instances

In terms of lateral movement within the intranet, we observe that the APT group OceanLotus exploits the CVE-2020-14882 WebLogic Remote Command Execution vulnerability, CVE-2021-22986 F5 BIG-IP iControl REST Unauthorized Remote Command Execution vulnerability, and JBoss deserialization vulnerability to roam within the intranet. Subsequently, the group creates a Powershell Webserver on servers at critical nodes within the intranet and stores encrypted CS payload on it.

Weakness Enumeration

CWE Entity Instances

CWE-ID	CWE Name	Source
CWE-918	Server-Side Request Forgery (SSRF)	NIST

CPAEC Entity Instances

CAPEC-ID	Attack Pattern Name
CAPEC-664	Server Side Request Forgery

IOC Categories: Md5

Vulnerability ID: CVE-2021-22986

Vulnerability Name: F5 BIG-IP and BIG-IQ Centralized Management iControl REST Remote Code Execution Vulnerability

Vulnerability Description: On BIG-IP versions 16.0.x before 16.0.1.1, 15.1.x before 15.1.2.1, 14.1.x before 14.1.4, 13.1.x before 13.1.3.6, and 12.1.x before 12.1.5.3 and BIG-IQ 7.1.0.x before 7.1.0.3 and 7.0.0.x before 7.0.0.2, the iControl REST interface has an unauthenticated remote command execution vulnerability. Note: Software versions which have reached End of Software Development (EoSd) are not evaluated.

Weakness ID: CWE-918

Weakness Name: Server-Side Request Forgery (SSRF)

Weakness Description: The web server receives a URL or similar request from an upstream component and retrieves the contents of this URL, but it does not sufficiently ensure that the request is being sent to the expected destination.

Attack Pattern ID: CAPEC-78

Attack Pattern Name: Server Side Request Forgery

Attack Pattern Description: An adversary exploits improper input validation by submitting maliciously crafted input to a target application running on a server, with the goal of forcing the server to make a request either to itself, to web services running in the server's internal network, or to external third parties. If successful, the adversary's request will be made with the server's privilege level, bypassing its authentication controls. This ultimately allows the adversary to access sensitive data, execute commands on the server's network, and make external requests with the stolen identity of the server. Server Side Request Forgery attacks differ from Cross Site Request Forgery attacks in that they target the server itself, whereas CSRF attacks exploit an insecure user authentication mechanism to perform unauthorized actions on the user's behalf.

Fig. 2. Entity extraction example.

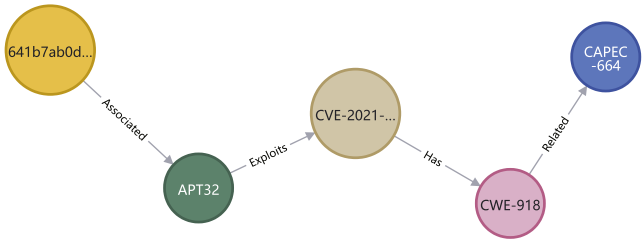


Fig. 3. Knowledge graph example of APT32 attack.

processing the extracted data, the data is concatenated based on the real-world relationships of APT attacks to construct the feature matrix, relationship matrix, and triples. The relationship matrix links the attack data that shares existing relationships, and this paper constructs a 420×2 relationship matrix. The features in the feature matrix are derived from the textual information of each entity. After removing irrelevant information, the frequently occurring words are identified as feature words, which are then used to build a feature dictionary. The feature matrix is constructed based on this feature dictionary, resulting in a 11424×134 matrix. As shown in Fig. 4, the following section will detail the steps of node feature extraction.

After obtaining the data, further processing is performed to obtain the feature matrix, adjacency matrix, and triples. The Algorithm 1 is presented below.

The node feature extraction in this paper consists of four main parts. The first part is reading the data and cleaning it. This involves removing unnecessary words from the text, such as 'a', 'an', 'the', and 'for', converting all text to lowercase, and tokenizing the data. The second part is constructing a feature dictionary. Words are selected based on their frequency of occurrence, and the necessary words are chosen from highest to lowest frequency according to the requirements to form the feature dictionary. Additionally, due to the specificity of IOCs, separate IOCs data are treated as individual features. The third part is extracting the relationships between nodes. Based on the information in the data, the relationships between each node are extracted for data construction.

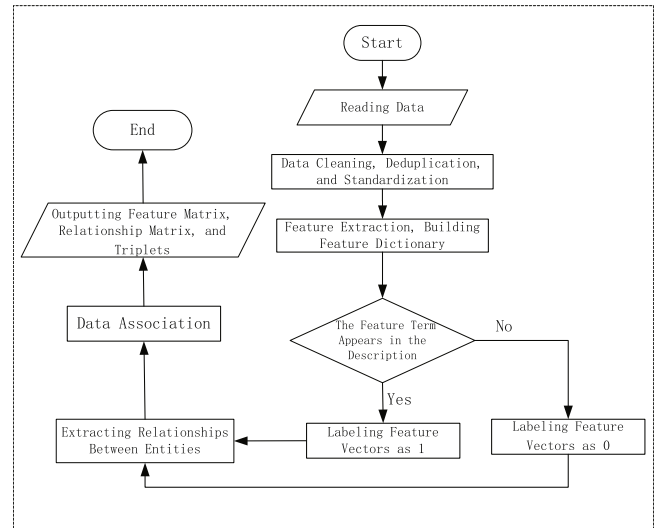


Fig. 4. Flowchart of node feature extraction.

Algorithm 1 Node feature extraction.

Input: the data **D**

Extract IOCs features

Construct the feature dictionary **F**

for D in F

if T words in D

 Print(1)

else

 Print(0)

end if

end for

for relation in R

 number each edge using an index dictionary

end for

Output: the feature matrix **X**, relationship matrix **A**, and triples **T**

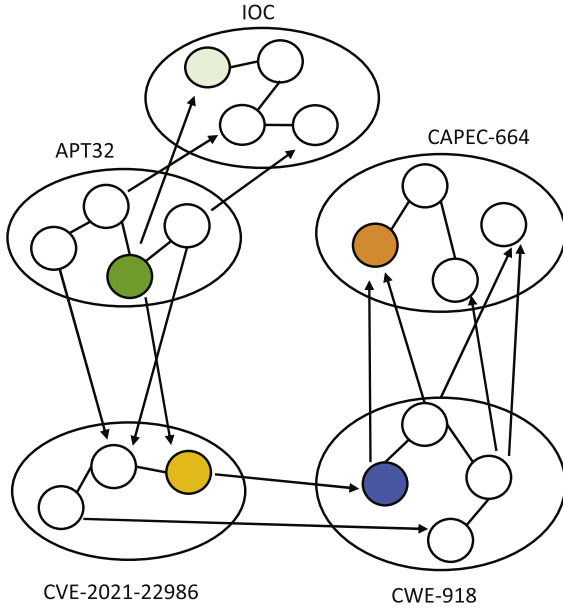


Fig. 5. Example of relationships between entities.

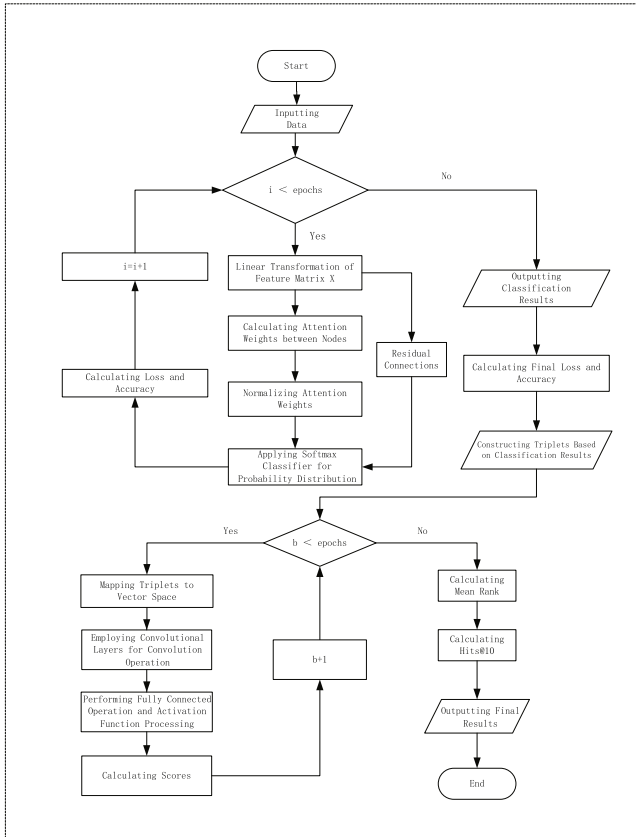


Fig. 6. Flowchart of GA-ConvE.

The fourth part involves constructing the feature matrix. Each part of the entity is matched to the constructed feature matrix, and the corresponding feature matrix is built. The final part is outputting the data. Based on the relationships between nodes, as shown in Fig. 5, the nodes are linked to form a complete attack chain. The feature matrices of each node are concatenated along the attack chain to construct the feature

matrix, and the extracted relationships are used to build triples for the adjacency matrix.

The construction of these relationships is based on identifying which IOCs tags were generated by the APT organization (APT32) during its attack on a specific institution and which Common Vulnerabilities and Exposures (CVE-2021-22986) were exploited. These vulnerabilities contain certain Common Weakness Enumerations (CWE-918), and based on these weaknesses, the corresponding attack methods and techniques (CAPEC-664) are selected. By treating CVE, CWE, CAPEC, and IOCs as nodes within a graph structure and constructing edges based on their relationships, a complete graph structure can be formed. In this graph, each node not only contains its own feature information but also has interconnections with the features of other nodes. The feature information possessed by each entity can effectively serve as an important supplement to the features of other entities. By constructing relationships between entities in this manner, the attack scenario carried out by an APT organization can be realistically recreated, thereby improving the accuracy of APT attack predictions.

5.2. APT attack prediction method based on GA-ConvE

The GA-ConvE APT attack prediction method first classifies APT attacks by their organization using the RMultiGAT model, and then uses the ConvE model to make targeted predictions on the classified data. This approach of classifying first and then predicting, combined with the advantages of both models, performs well in APT attack prediction. As shown in Fig. 6, the main process of APT attack prediction is illustrated.

The processed data is fed into the RMultiGAT model, where linear transformations, attention calculations, and normalization are performed. Finally, the classification results are obtained using the Softmax function. The triples constructed based on the classification results are then input into the ConvE model, where functions such as embedding and convolution are applied to derive the prediction results. The Algorithm 2 is as follows. The GA-ConvE method first classifies APT

Algorithm 2 GA-ConvE.

Input: the feature matrix X , relationship matrix A , and triples T
GAT(A, X)
 Linear Transformation of X and Computation of Node Weights to Obtain $e_{ij}^{(l)}$
 Normalizing Node Weights to Obtain $\alpha_{ij}^{(l)}$
 Using Activation Function to Compute Next Layer Nodes $h_j^{(l+1)}$ based on $\alpha_{ij}^{(l)}$
 Merging Attention Weights Weightedly to Obtain h_i'
 Calculating Classification Results using Softmax Function
 Calculating Accuracy and Loss
end for
Output: Final Classification Results
Classifying T According to the Classification Results
ConvE(T)
 The Embedding layer transforms the input matrix T into an embedding matrix E
 Performing Convolution Operation on E to Obtain Vector F
 Performing Fully Connected Operation and Non-linear ReLU Activation on F
 Calculating Scores
end for
 Calculating MR and Hits@10
Output: Final Prediction Results

attacks using a multi-layer Graph Attention Network with residual connections, and then predicts the classified attack data using a two-dimensional convolutional link prediction model. The multi-layer Graph

Attention Network with residual connections introduces an attention mechanism into the traditional convolutional neural network, allowing each node to dynamically assign weights based on its relationships with neighboring nodes. This enables flexible weight adjustments during information propagation. Additionally, the multi-head attention mechanism enhances the model's expressive power, allowing multiple sets of attention weights to be learned simultaneously, which better captures relationships in complex graph structures. The residual connections and layer attention allow the model to better learn node associations and improve its stability. The two-dimensional convolutional link prediction model, by incorporating convolution operations, effectively captures local patterns between entities and relationships. It formalizes the interactions between entities and relationships, introduces positional embeddings to handle symmetry, and automatically learns features. This model is not only suitable for triple classification tasks but also excels in link prediction tasks, demonstrating robustness and generality.

In the APT attack classification task, the model utilizes a weight matrix W to perform a linear transformation on the feature matrix X . The calculation formula is:

$$h_i^{(l)} = W^{(l)} \cdot x_i \quad (1)$$

Here, x_i is the feature vector of node i in the feature matrix X , and $h_i^{(l)}$ is the feature representation of node i after the linear transformation in the l th layer. By using linear combinations, the original APT attack features are transformed into new features that are more suitable for classification tasks. Then, the *LeakyReLU* function is used to compute the attention weights between the nodes after the linear transformation. The calculation formula is:

$$e_{ij}^{(l)} = \text{LeakyReLU}\left(a^{(l)T} [W^{(l)} \cdot h_i^{(l)} \parallel W^{(l)} \cdot h_j^{(l)}]\right) \quad (2)$$

Here, $a^{(l)T}$ represents the attention weight vector at layer l , $W^{(l)}$ is the weight matrix at layer l , $h_i^{(l)}$ is the feature representation of node i at layer l , and $h_j^{(l)}$ is the feature representation of node j at layer l . In this paper, the main focus is on calculating the similarity and correlation between APT attack nodes using the activation function. The computed weights between the nodes need to be normalized. The normalization formula is:

$$\alpha_{ij}^{(l)} = \frac{\exp(e_{ij}^{(l)})}{\sum_{k \in N(i)} \exp(e_{ik}^{(l)})} \quad (3)$$

Here, $\alpha_{ij}^{(l)}$ is the attention weight allocated by node i to node j at layer l , and $N(i)$ represents the set of neighboring nodes of APT attack node i . In this paper, the primary focus is on dynamically adjusting the attention paid to neighboring APT attack nodes based on the local graph structure and features of each APT attack node. For each node in the dataset, the computed attention weights are aggregated using a weighted sum, and the calculation formula is:

$$h_i' = \frac{1}{M} \sum_{m=1}^M h_i^{(m)} \quad (4)$$

Here, $h_i^{(m)}$ is the weight of node i at layer m , and h_i' is the final representation of node i . The input feature vector and the computed feature are then summed via residual connections to obtain the final output feature. The calculation formula is:

$$h_i^{\text{out}} = h_i^{(l)} + h_i' \quad (5)$$

Finally, the Softmax function is applied to all the output nodes to compute the classification prediction values, and the highest value is selected as the classification result. In this way, the model transforms the output of APT attack data into interpretable classification results.

The constructed triples T are classified according to the APT attack classification results, and the classification results can then be used to make targeted predictions for APT attacks. In the APT prediction task, an embedding layer is used to map the input triples into continuous

vector space, and the calculation formula is:

$$E = [e_h; e_r] \quad (6)$$

Here, e_h and e_r are the embedding vectors of the head entities (IOCs, APT, CVE, etc.) and the relationships (Associated, Exploits, Has, etc.), respectively. These embedding vectors represent the spatial positions of these security entities and relationships, capturing their semantic and relational information. After embedding, convolution operations are performed on them using a convolutional layer. The calculation formula is:

$$F_i = \text{ReLU}\left(\sum_{j=1}^k \sum_{l=1}^d \omega_{ijl} \cdot E_{(i+l-1),j}\right) \quad (7)$$

Here, F_i is the i th element of the output from the convolutional layer, ω_{ijl} represents the parameters of the convolution kernel, k is the size of the convolution kernel, and d is the dimension of the embedding layer. The purpose is to extract local feature patterns from the embedded vectors through convolution operations, which are used to identify relationships and patterns between different features. After the convolution, the output is processed through a fully connected layer and a non-linear activation function to obtain the final mapping result. The calculation formula is:

$$M = \text{ReLU}(W_m \cdot F + b_m) \quad (8)$$

Here, W_m is the weight matrix of the fully connected layer, and b_m is the bias vector. This step allows the integration of multiple security entity features for deeper reasoning. Finally, a scoring function is used to compute the final prediction score. The calculation formula is:

$$\text{Score} = \sigma(W_s \cdot M + b_s) \quad (9)$$

Here, W_s is the weight matrix of the final fully connected layer, b_s is the bias vector, and σ is the *sigmoid* activation function. The output is mapped to the range (0, 1), and based on the prediction score, the APT attack is predicted.

By first classifying APT attacks using a multi-layer Graph Attention Network with residual connections, the model enhances its deep understanding of network topology structures and effectively captures multi-level attack features. Subsequently, the two-dimensional convolutional link prediction model is employed to predict the classified attack data, further enhancing the model's perception of semantic information within the knowledge graph. As illustrated in Fig. 7, this classify-first-then-predict strategy fully leverages the strengths of both models, enabling the identification of APT attack types as well as more accurate predictions of attack behaviors.

5.3. Algorithmic complexity analysis

(1) Classification Model Analysis

The computational cost of RMultiGAT primarily comes from neighbor aggregation and attention weight computation in the multi-head attention mechanism. Given N nodes, E edges, input feature dimension F , output dimension F' , K attention heads, and L layers, the time complexity of a single layer is $O(NFF' + EK F')$, and stacking L layers yields $O(L(NFF' + EK F'))$. In sparse graphs where $E \approx O(N)$, the complexity grows nearly linearly with graph size, ensuring scalability. The space complexity is dominated by storing node embeddings and attention weights, i.e., $O(NF' + EK)$, while residual connections require additional memory for intermediate embeddings of order $O(NF')$. Compared with GCN, whose complexity is $O(L(NFF' + EF'))$, RMultiGAT introduces an extra factor K due to multi-head attention, slightly increasing computation. Nevertheless, the richer representational capacity and improved stability from residual connections make RMultiGAT more effective than GCN, GIN, or GraphSAGE for complex classification tasks.

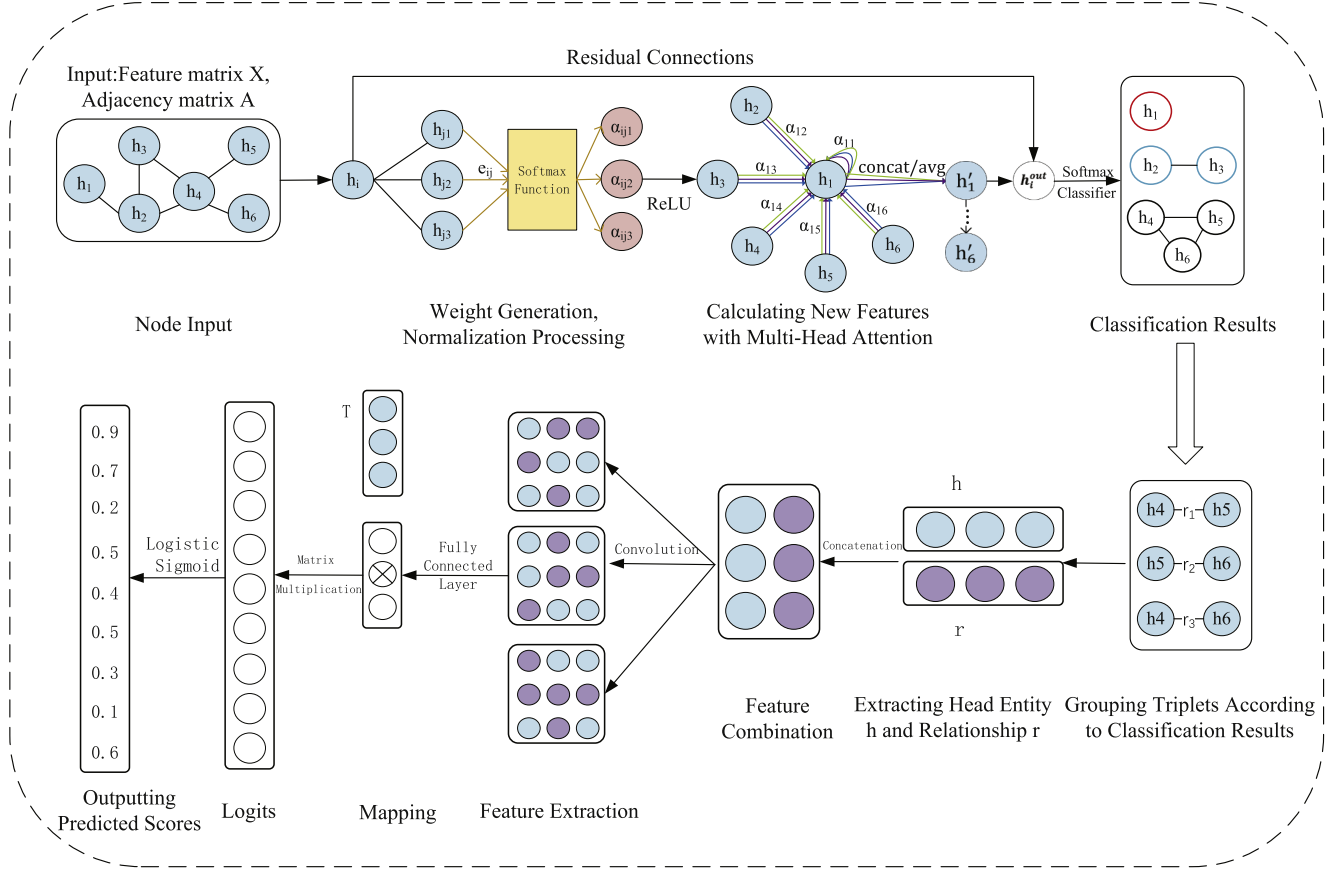


Fig. 7. Schematic diagram of GA-ConvE.

(2) Prediction Model Analysis

The computational cost of ConvE arises mainly from embedding transformation, two-dimensional convolution, and fully connected projection. Given $|E|$ entities, $|R|$ relations, embedding dimension d , kernel size $k \times k$, number of filters c , and batch size B , the per-iteration time complexity is $O(B(ck^2d + |E|d))$. The space complexity includes storing entity and relation embeddings $O(|E|d + |R|d)$, together with convolution kernels and fully connected parameters $O(ck^2 + d|E|)$. Compared with lightweight models such as TransE and DistMult/ComplEx, which require only $O(Bd)$, ConvE incurs higher training cost due to the convolutional component; however, the convolution operation enhances the model's ability to capture local semantic patterns and entity–relation interactions, thereby providing superior predictive accuracy and interpretability relative to simpler embedding models.

6. Experimental simulation and analysis

In the experiments, we employed a Windows 10 operating system with 16 GB of memory, a 2.2GHz six-core Intel Core i7 CPU, and a GTX 1050 GPU. The models were trained using PyTorch 2.0.1, TensorFlow 1.15.5, and Python 3.7. Furthermore, we compared the classification performance of RMultiGAT, GAT, GraphSAGE, GIN, GCN, DGCNN, and GGNN, as well as the prediction results from five models, namely ConvE, TransE, ComplEx, DistMult, and HolE. To evaluate the effectiveness of two-dimensional convolution, one-dimensional convolution was used as a substitute for comparison. The classification models were implemented with GPU acceleration, while the prediction models were executed on CPU. In addition, the proposed method was compared with previously introduced graph neural network-based approaches and validated through experiments conducted in real-world scenarios.

6.1. Dataset

In this study, we primarily collected representative APT reports and extracted the required key information, as shown in Table 2. This includes details such as the organization responsible for the attack and the vulnerabilities exploited. A knowledge graph and triples were then constructed by associating relevant security entities. The descriptions of CVE, CWE, and CAPEC were obtained from MITRE, while the IOC labels were mainly derived from the APT reports. The dataset incorporates APT attack data from 12 organizations. Due to the limited availability of real attack data, we constructed a total of 11,424 data points for classification during data collection, with 2856 data points representing actual attack data, and the remainder representing non-attack data. Additionally, from these 2856 real attack data points, 11,424 triples were constructed for use in attack inference and prediction.

In the experiment, we employed the RMultiGAT model and the ConvE model for pre-training, aiming to prepare for APT attack organization classification and APT attack prediction. For the APT attack classification experiment, we applied cross-validation to process the dataset, splitting it into multiple subsets. The dataset was divided into training, testing, and validation sets in an 8:1:1 ratio to enhance the accuracy of the experiment. In the APT attack prediction experiment, we randomly selected 20% of the data from each of the 12 types of attack data to form 12 test sets. Each group of data was trained individually, and when comparing models, we averaged the results from the 12 groups to conduct the comparison.

6.2. Evaluation indicators

Evaluation of classification results includes accuracy, loss rate, precision, recall, and F1-score. Additionally, average rank (Mean Rank) and

Table 2
Entity description in the dataset.

Entity Type	Entity Example	Entity Description
APT	APT-C-32, APT-C-35, APT-Q-40, APT-C-55, APT-C-17, APT-C-08, APT37, APT-C-09	APT refers to a sophisticated and organized form of cyberattack, typically carried out by state-sponsored or highly advanced hacker groups.
CVE	CVE-2018-20250, CVE-2018-8453, CVE-2020-0688, CVE-2021-21551, CVE-2021-26857, CVE-2022-27925	CVE is a public database maintained by the MITRE organization, used to assign unique identifiers to known cybersecurity vulnerabilities.
CWE	CWE-119, CWE-120, CWE-125, CWE-131, CWE-20, CWE-200, CWE-732	CWE is a resource for software design and implementation flaws (weaknesses), describing common types of errors.
CAPEC	CAPEC-1, CAPEC-10, CAPEC-101, CAPEC-108, CAPEC-109, CAPEC-123	CAPEC is a database of known attack methods and patterns that describes how attackers exploit vulnerabilities and weaknesses to carry out attacks.
IOC	domain: alphacrane.com, ipv4: 81.95.7.12, url: s2.cdn-edge.net/cart.php	IOC refers to evidence used to identify whether a system or network has been compromised or is under attack.

hit rate at top-n (HITS@n) are used for prediction evaluation, with n set to 10 in this experiment.

It is important to note that in the classification experiment, these metrics are calculated using TP (true positive instances), FP (false positive instances), TN (true negative instances), and FN (false negative instances). However, due to the dataset being a multi-classification dataset related to APT attacks, the calculation method of these metrics has changed compared to binary classification data. In this paper, TP refers to the number of instances correctly classified for a specific category, FP indicates the number of instances in this category mistakenly classified as other organizational categories, TN represents the total number of samples of other classes when classifying data into a specific group, and FN indicates the number of instances of other categories classified as this specific category. The following will provide a detailed explanation of these metrics.

Accuracy is the simplest and most intuitive evaluation metric in the GAT model. It measures the proportion of correct predictions in the classification of APT attack data. The higher the accuracy, the better. Its calculation formula is as follows:

$$accuracy = \frac{TP + TN}{TP + FP + TN + FN} \quad (10)$$

The loss rate is determined by the cross-entropy loss used during the training process, and a smaller value is preferable.

Precision and recall are a pair of interrelated metrics. Precision measures the proportion of all APT attack data predicted as a specific APT class that truly belongs to that class. Recall measures, among all samples of a particular APT attack class, the proportion correctly predicted as that class. A higher recall value is desirable. The calculation formulas are as follows:

$$precision = \frac{TP}{TP + FP} \quad recall = \frac{TP}{TP + FN} \quad (11)$$

The F1-score is a metric that comprehensively considers both precision and recall. It is used to balance the relationship between the two metrics, with a higher value being preferable. The calculation formula is as follows:

$$F1 - score = 2 \times \frac{precision \times recall}{precision + recall} \quad (12)$$

The mean rank is calculated for each test triplet, where the model ranks all possible tail entities based on their predicted scores. The actual rank of the tail entity is then computed, and a smaller value is preferable. The calculation formula is as follows:

$$Mr = \frac{1}{N} \sum_{i=1}^N Rank_i \quad (13)$$

The Hits@n rate is calculated for each test triplet by checking if the correct tail entity is among the top n pre-ranked entities by the model. If it is, it is considered a "hit," and a smaller value is preferable. The

calculation formula is as follows:

$$Hits@n = \frac{\text{Number of Hits}}{N} \quad (14)$$

6.3. Parametric experiments

In this section, we consider the impact of hyperparameters on the model. In this experiment, we analyze the effects of six hyperparameters in the RMultiGAT model-lr (learning rate), hidden (number of hidden layers), epochs (number of iterations), dropout (dropout rate), layers (number of attention layers), and nheads (number of attention heads)-using three metrics: accuracy, loss, and F1. For the ConvE model, we analyze the effects of three hyperparameters-batches_count (number of batches), epochs (number of iterations), and k (embedding dimension)-using the Mr and Hits@10 metrics.

The selection of hyperparameters is crucial for model training and performance. The learning rate directly affects the step size for parameter updates, the hidden layer dimension determines the representation dimension of the nodes, and the number of attention heads enhances the model's expressive capability. The dropout rate is used to prevent overfitting, while increasing or decreasing the number of attention layers tests the impact of model complexity on performance. The number of epochs affects how many times the model learns from the data, and the batch size balances learning speed with generalization ability. An appropriate embedding dimension helps the model better capture data patterns. Adjusting these parameters optimally can enhance model performance, but they require validation and fine-tuning in experiments to find the best hyperparameter combination for a specific task and dataset. Experiments tested hyperparameters for both classification and prediction models, resulting in the findings shown in Figs. 8 and 9.

This experiment iteratively searched for the optimal hyperparameter combination using nested loops and performed parameter experiments on individual hyperparameters to analyze their effects on the model. As shown in Fig. 8, the parameter experiments for the GAT model are illustrated. The results in Fig. 8(a) show the impact of different parameters for nheads (2, 4, 8, 16, 32). It can be observed that as the number of attention heads increases, the model's accuracy and F1 score also increase, reaching their peak when the number of heads is 8. Beyond this point, increasing the number of heads does not significantly improve performance; in fact, performance either slightly decreases or remains stable. The conclusion is that when nheads equals 8, the model performs best. Larger attention head numbers help capture complex graph structures, but the performance gains diminish as the number increases, ultimately reducing model performance. In Fig. 8(b), the results of varying the number of attention layers (layers: 1, 2, 3, 4, 5) are shown. As the number of layers increases from 1 to 2, the model's performance improves, reaching its best performance. However, further increases in the number of layers lead to a decline in performance. The conclusion is that an appropriate number of attention layers can enhance the model, but

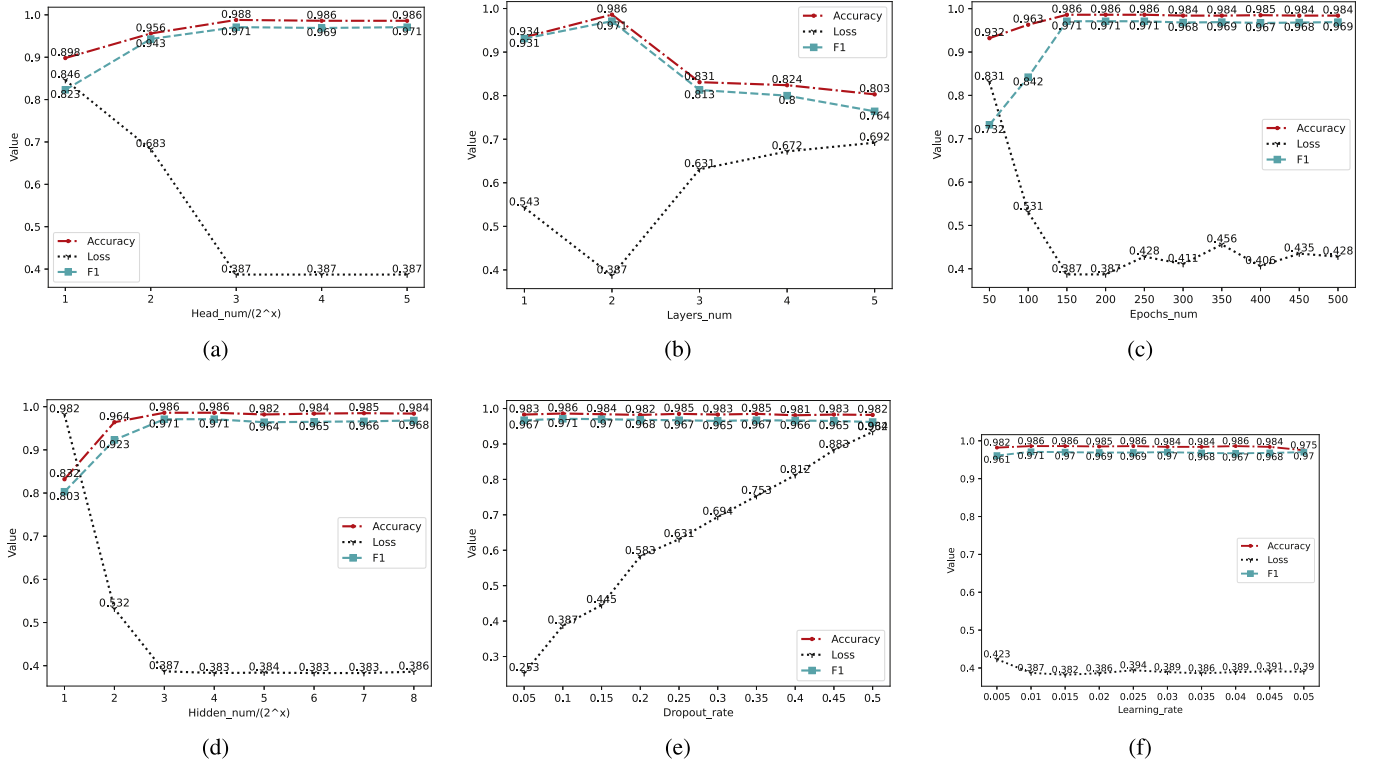


Fig. 8. Hyperparameter experimental graph for eRMultiGAT model (a) attention heads (b) hidden layer (c) epochs (d) hidden layers (e) dropout rate (f) learning rate.

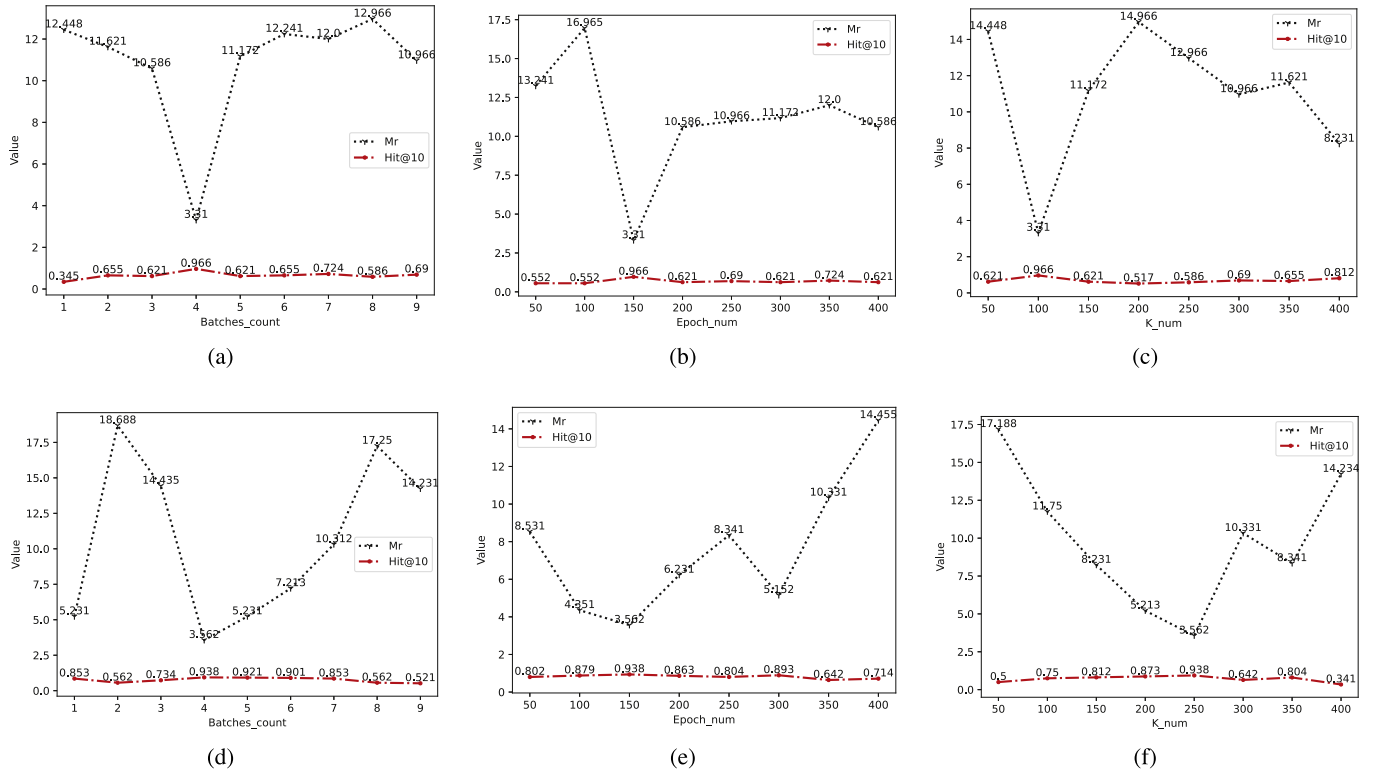


Fig. 9. Hyperparameter experimental graph for the ConvE model APT-C-09: (a) batches_count (b) epochs (c) embedding dimension APT-C-26: (d) batches_count (e) epochs (f) embedding dimension.

Table 3
Hyperparameters for the e RMultiGAT model.

Parameter	Value	Parameter	Value
lr	0.01	nheads	8
hidden	8	dropout	0.1
epochs	150	layers	2
weight_decay	5e-4	seed	42

too many layers negatively impact performance. In Fig. 8(c), the analysis of varying the number of epochs (epochs: 50, 100, 150, 200, 250, 300, 350, 400, 450, 500) shows that accuracy and F1 scores change significantly between 50 and 150 epochs, reaching optimal performance. Beyond 150 epochs, the model's performance does not improve further and even starts to decline. The conclusion is that a suitable number of training epochs can effectively improve model performance, while too many epochs may lead to overfitting without further performance gains. In Fig. 8(d), the impact of varying hidden layer dimensions (hidden: 2, 4, 8, 16, 32, 64, 128, 256) on the model is shown. As the number of hidden layers increases, the model's performance improves, with accuracy and F1 scores reaching their highest values at hidden = 8. Beyond this point, the values fluctuate within a certain range. The conclusion is that appropriate hidden layer dimensions enhance model performance, but increasing the dimensions too much may not result in further improvements. In Fig. 8(e), the analysis of different dropout rates (dropout: 0.05, 0.1, 0.15, 0.2, 0.25, 0.3, 0.35, 0.4, 0.45, 0.5) shows that the dropout rate has a significant impact on loss. As the dropout value increases, the loss also increases. The conclusion is that when dropout equals 0.1, the model performs best. An appropriate dropout rate enhances the model's generalization ability, while too large a dropout rate leads to underfitting. In Fig. 8(f), the analysis of different learning rates (lr: 0.005, 0.01, 0.015, 0.02, 0.025, 0.03, 0.035, 0.04, 0.045, 0.05) shows that a learning rate that is too low cannot fully exploit the model's potential. The conclusion is that when lr equals 0.01, the model performs optimally. Only an appropriate learning rate can accelerate convergence and improve model performance.

The results of this experiment indicate that when the number of attention heads is set to 8, the number of attention layers is set to 2, the number of iterations (epochs) is set to 150, the number of hidden layers is set to 8, the dropout rate is set to 0.1, and the learning rate is set to 0.01, the model achieves the best training results. Therefore, the optimal parameters were used for this experiment, resulting in the best classification performance. The hyperparameter settings for the model are shown in Table 3.

Using the RMultiGAT classification model, APT attacks were divided into 12 categories. Two of these categories (APT-C-09 and APT-C-26) were selected for hyperparameter experiments. These two were chosen because APT-C-09 and APT-C-26 represent different attack strategies and behavior patterns, making them both typical and representative. APT-C-09 likely represents a more traditional attack method with certain regularities and patterns, while APT-C-26 may involve more complex and diverse attack techniques. This difference makes them ideal choices for testing model performance, especially when evaluating the model's ability to handle diverse attack behaviors. Additionally, these two organizations are more active and pose greater threats compared to other APT organizations. As shown in Fig. 9, the results of setting different parameters for batches_count (1, 2, 3, 4, 5, 6, 7, 8, 9) are presented in Fig. 9(a) and (d). The results indicate that the batch count has a significant effect on model performance, with both datasets showing optimal results when the batch count is 4. The Hits@10 values do not vary significantly across different batch counts, suggesting that the batch size has a smaller impact on model accuracy but a larger impact on model stability. By analyzing the effects of different epochs (50, 100, 150, 200, 250, 300, 350, 400) on the reasoning model, the results are shown in Fig. 9(b) and (e). It can be concluded that the number of iterations primarily affects the fluctuation of the Mr value. Both datasets achieve the

Table 4
Hyperparameters for the ConvE model.

APT organizations	batches_count	k	epochs	Mr	Hits@10
APT-C-32	2	100	250	1.34	1
APT-C-35	1	150	100	1.14	1
APT-Q-40	2	250	100	1.54	1
APT-C-55	1	200	300	1.34	1
APT-C-17	2	100	200	1.26	1
APT-C-08	4	200	300	1.633	1
APT-C-01	4	200	300	5.02	0.816
APT32	4	100	150	3.93	0.843
APT-C-26	4	150	250	3.562	0.938
APT-C-06	2	300	200	6.54	0.83
APT37	1	150	200	4.43	0.926
APT-C-09	4	150	100	3.31	0.996

Table 5
Comparison of GA-ConvE predictive effects.

Model	Mr	Hits@10
GA-ConvE	2.920	0.946
DP-ConvE	5.014	0.851
EM-GA-ConvE	2.891	0.956

best results when the number of epochs is 150. Fewer iterations may result in insufficient learning, while too many iterations may lead to overfitting. Finally, by analyzing the effects of different k values (50, 100, 150, 200, 250, 300, 350, 400) on the model, the results are shown in Fig. 9(c) and (f). It can be observed that both datasets achieve optimal performance when the embedding dimensions are set to 100 and 250, respectively. appropriate parameters can optimize the model's performance, but increasing the parameters excessively may lead to overfitting.

Based on the above experiments, it can be observed that the optimal hyperparameter combinations for different datasets and models vary. By utilizing nested loops for hyperparameters, the optimal hyperparameter combinations for the twelve categories of data were determined. The model's hyperparameter settings are shown in Table 4.

6.4. Comparative tests

This paper proposes the GA-ConvE method, which first classifies attack data and then predicts the classified APT attacks based on the classification results. To demonstrate the superiority and robustness of the proposed method, this section compares the average prediction results of the GA-ConvE method with the results of direct prediction (DP-ConvE). Since the classification results are not 100 % accurate due to the limitations of the classification model, this section also compares the average prediction results of the GA-ConvE method with the average prediction results after removing misclassified items (EM-GA-ConvE). The experimental results are shown in Table 5.

By comparing the Mr and Hits@10 of GA-ConvE and DP-ConvE, it can be observed that the predictive performance is significantly improved using the GA-ConvE method, demonstrating its superiority. Further comparison between GA-ConvE and EM-GA-ConvE in terms of Mr and Hits@10 indicates that, although the classification results are not perfect due to the influence of the classification model, the impact of misclassifications can be negligible. Combining these results with DP-ConvE, it is evident that the GA-ConvE method greatly enhances the predictive effectiveness of APT attacks. Moreover, it is feasible in real-world scenarios.

Compared to traditional classification models, the RMultiGAT model introduces a multi-layer attention network and connects the layers through residual connections, which significantly enhances the model's performance. Additionally, the multi-head attention mechanism improves the model's expressive power by allowing the simultaneous

Table 6
Classification model comparison.

Model	Precision	Recall	F1 Score	Loss	Accuracy	Time/(s)	GPU/(GB)
GraphSAGE	0.738	0.754	0.733	0.626	0.902	20.5	2.53
GIN	0.930	0.702	0.685	0.190	0.944	30.4	2.69
GCN	0.841	0.827	0.828	0.447	0.944	26	2.94
DGCNN	0.833	0.889	0.852	0.154	0.889	35.5	2.74
GGNN	0.961	0.969	0.963	0.137	0.978	44.2	2.85
GAT	0.860	0.857	0.856	0.496	0.950	54.9	3.42
RMultiGAT	0.972	0.970	0.971	0.387	0.986	58.1	3.74

learning of multiple sets of different attention weights, thereby better capturing relationships in complex graph structures. In this paper, the RMultiGAT model is compared with several other models, including GAT, GraphSAGE, GIN, GCN, DGCNN, and GGNN. The experimental results are shown in Table 6. Although the proposed model exhibits slightly higher runtime and memory consumption compared with other approaches, it demonstrates significantly superior classification performance. Among the five evaluation metrics, four achieve the best results, which fully highlights the effectiveness of the model. This outcome indicates that, by incorporating multi-layer residual attention structures and multi-head mechanisms, the model can better capture the complex graph-structural relationships and latent semantic information inherent in APT attack data, thereby achieving comprehensive improvements in key metrics such as precision, recall, and F1-score. While runtime and memory overheads increase to some extent, these costs remain within an acceptable range. Experimental results further suggest that the model can still meet computational resource requirements in real-world deployment scenarios, and the performance gains in classification provide substantial practical value for security defense and threat detection. In other words, when dealing with APT attacks characterized by high complexity and diversity, moderately increasing computational expenditure in exchange for higher classification accuracy and stability represents a reasonable and necessary trade-off.

To compare the superiority of the ConvE model relative to other models, APT attack data was first classified using the RMultiGAT model. The classified data was then sequentially input into the ConvE model according to their categories, and the prediction performance of the models was compared. The classification results of the attack data are shown in Fig. 10. As can be seen from the figure, the RMultiGAT model divided the attack data into 12 categories. In the APT-C-06 category, 432 attack data points were correctly classified, but 4 attacks were misclassified into the APT-C-09 category. In the APT-C-09 category, 439 attack data points were correctly classified, but 1 attack was misclassified into the APT-C-26 category. Other results follow a similar pattern.

The classified data was input into each prediction model, and the experimental results are shown in Table 7. From the table, it can be observed that the Mr metric for the ComplEx and HolE models is smaller than that of the ConvE model for the APT-C-32 category. However, in the APT-C-08 category, the ConvE model has a smaller Mr value compared to the other four models. This shows that it is not possible to determine the superiority of a model based solely on the prediction results of one data category. Therefore, the average prediction results across all 12 attack categories were calculated. As seen from the table, the ConvE model's prediction results outperform those of the other models.

To avoid the interference of "optimized parameters" in complexity evaluation, all predictive models in this study were compared under identical hardware and software environments with uniform, non-tuned hyperparameter configurations. Under these consistent settings, the experimental results are presented in Table 8. The average training time of ConvE is slightly higher than that of the baseline methods, with ConvE incurring greater runtime overhead compared with TransE, ComplEx, DistMult, and HolE. This increase is primarily attributed to the additional computational cost introduced by two-dimensional convolution during relation modeling, which raises the per-iteration computa-

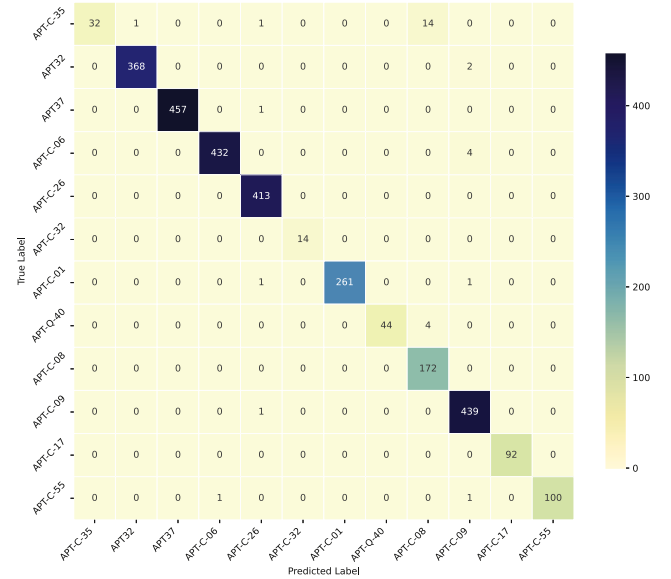


Fig. 10. APT attack classification results.

tional complexity. However, in terms of memory consumption, ConvE demonstrates a certain advantage, requiring less storage overhead than some of the comparative models. These results indicate that, although ConvE sacrifices some time efficiency, it offers improved memory efficiency, which is of practical significance for handling large-scale graph data.

To demonstrate the role of two-dimensional convolution in the GA-ConvE method, this paper also replaced the two-dimensional convolution with one-dimensional convolution (ConvKB) to further evaluate the effectiveness of the two-dimensional convolution in this method. The prediction was performed on data from the twelve attack organizations, and the results were averaged. The experimental results are shown in Fig. 11. From the results, it can be seen that the Mr and Hits@10 metrics obtained using one-dimensional convolution are both worse than those achieved with two-dimensional convolution, clearly demonstrating the superiority of two-dimensional convolution in this method.

To further demonstrate the performance of the GA-ConvE model, this approach was compared with the original GAT method and other graph neural network-based methods. The 12 categories of attack data extracted in this paper were used for prediction, and evaluation metrics such as Precision, Recall, and F1-Score were used to assess these methods. The experimental results are shown in Table 9. From the table, it can be seen that for five of the twelve attack categories, the method proposed in this paper achieved the best results, and overall, it outperformed the other methods.

6.5. Application experiment

To more comprehensively reveal the full scope of APT attacks and thereby defend against them more effectively, this paper proposes the GA-ConvE prediction method. By using a classify-then-predict approach,

Table 7
Performance comparison of prediction models.

	ConvE		TransE		ComplEx		DistMult		HolE	
	Mr	Hits@10	Mr	Hits@10	Mr	Hits@10	Mr	Hits@10	Mr	Hits@10
APT-C-32	1.340	1.000	1.000	1.000	1.240	1.000	1.150	1.000	1.000	1.000
APT-C-35	1.140	1.000	3.157	1.000	3.340	1.000	3.230	1.000	3.430	1.000
APT-Q-40	1.540	1.000	1.355	1.000	1.355	1.000	1.642	1.000	2.232	1.000
APT-C-55	1.340	1.000	1.753	1.000	1.130	1.000	1.643	1.000	1.750	1.000
APT-C-17	1.260	1.000	1.389	1.000	1.532	1.000	1.243	1.000	1.875	1.000
APT-C-08	1.633	1.000	2.211	1.000	2.880	0.920	2.424	1.000	2.431	0.980
APT-C-01	5.020	0.816	4.911	0.863	5.960	0.890	6.140	0.834	5.742	0.806
APT32	3.930	0.843	7.763	0.843	9.840	0.830	8.432	0.742	7.643	0.786
APT-C-26	3.562	0.938	3.563	0.950	4.130	0.880	3.242	0.884	4.214	0.914
APT-C-06	6.540	0.830	8.553	0.707	7.230	0.790	7.421	0.785	7.562	0.810
APT37	4.430	0.926	4.658	0.813	3.313	0.955	5.740	0.854	5.454	0.850
APT-C-09	3.310	0.996	3.346	0.871	4.755	0.898	4.023	0.913	4.035	0.879
Average	2.920	0.946	3.638	0.921	3.892	0.930	3.861	0.918	3.947	0.919

Table 8
Complexity comparison of prediction models.

	ConvE		TransE		ComplEx		DistMult		HolE	
	Time (/s)	RSS (/MB)	Time (/s)	RSS (/MB)	Time (/s)	RSS (/MB)	Time (/s)	RSS (/MB)	Time (/s)	RSS (/MB)
APT-C-32	5.8	312.6	5.8	306.9	5.9	318.6	5.7	309.9	5.9	337.2
APT-C-35	6.2	331.5	5.6	336.6	5.7	340.2	5.7	337.2	5.6	328.5
APT-Q-40	6.5	325.2	5.9	330.9	5.6	329.4	6.0	329.1	6.1	339.9
APT-C-55	6.9	347.4	6.4	343.5	6.3	353.4	6.2	354.9	6.4	348.3
APT-C-17	7.6	365.1	6.3	370.2	6.2	376.8	6.6	357.9	6.9	356.7
APT-C-08	7.4	373.8	6.8	379.5	6.6	397.5	6.7	371.1	6.8	381.6
APT-C-01	11.3	134.8	8.3	410.4	8.5	431.1	8.8	406.2	10.3	404.1
APT32	11.7	421.5	9.5	418.5	9.3	456.6	9.5	419.1	10.1	410.4
APT-C-26	15.6	517.2	10.9	504.9	11.9	555.3	11.8	508.5	13.2	513.6
APT-C-06	13.3	458.4	11.7	462.3	10.8	500.1	11.2	460.2	12.5	454.2
APT37	9.8	408.9	10.6	407.4	10.4	429.6	10.6	411.9	11.6	408.3
APT-C-09	13.5	459.6	10.5	462.9	10.8	502.8	10.9	463.5	12.7	456.3
Average	9.63	371.33	7.98	394.50	8.34	415.95	8.31	394.13	9.01	394.93

Table 9
Comparison of GA-ConvE with other methods.

	Duan et al. (2023)			Tang et al. (2023)			Ren et al. (2023)		
	Precision	Recall	F1-Score	Precision	Recall	F1-Score	Precision	Recall	F1-Score
APT-C-32	1.000	0.929	0.963	1.000	0.857	0.923	1.000	0.857	0.923
APT-C-35	1.000	0.479	0.648	0.919	0.992	0.954	0.583	0.292	0.389
APT-Q-40	0.797	0.979	0.879	1.000	0.479	0.648	1.000	0.313	0.476
APT-C-55	1.000	0.980	0.990	0.505	0.980	0.667	1.000	0.980	0.990
APT-C-17	1.000	1.000	1.000	1.000	0.652	0.789	0.919	0.989	0.953
APT-C-08	0.930	1.000	0.964	0.715	0.744	0.729	0.771	1.000	0.871
APT-C-01	0.919	0.992	0.954	1.000	1.000	1.000	1.000	0.992	0.996
APT32	0.997	0.995	0.996	0.989	1.000	0.995	0.992	0.997	0.995
APT-C-26	0.992	0.956	0.974	1.000	1.000	1.000	0.993	1.000	0.996
APT-C-06	0.993	0.995	0.994	1.000	1.000	1.000	0.998	0.998	0.998
APT37	1.000	0.998	0.999	1.000	0.993	0.997	1.000	1.000	1.000
APT-C-09	0.950	0.998	0.973	1.000	1.000	1.000	0.989	0.995	0.992
Average	0.965	0.942	0.944	0.927	0.892	0.892	0.937	0.868	0.882

	Zhao et al. (2024)			Original			Ours		
	Precision	Recall	F1-Score	Precision	Recall	F1-Score	Precision	Recall	F1-Score
APT-C-32	0.797	0.979	0.879	1.000	0.857	0.923	1.000	1.000	1.000
APT-C-35	1.000	0.042	0.080	1.000	0.500	0.667	1.000	0.667	0.800
APT-Q-40	1.000	0.833	0.909	0.957	0.938	0.947	1.000	0.917	0.957
APT-C-55	1.000	0.706	0.828	1.000	0.980	0.990	1.000	0.980	0.990
APT-C-17	1.000	1.000	1.000	0.820	0.989	0.897	1.000	1.000	1.000
APT-C-08	0.781	0.994	0.875	0.988	0.994	0.991	0.905	1.000	0.950
APT-C-01	0.970	0.996	0.983	1.000	0.992	0.996	1.000	0.992	0.996
APT32	1.000	0.995	0.997	0.992	0.997	0.995	0.997	0.995	0.996
APT-C-26	0.998	1.000	0.999	0.995	0.998	0.996	0.990	1.000	0.995
APT-C-06	0.977	0.986	0.982	0.998	0.998	0.998	0.998	0.991	0.994
APT37	1.000	0.998	0.999	1.000	0.993	0.997	1.000	0.998	0.999
APT-C-09	0.905	0.991	0.946	0.976	1.000	0.988	0.982	0.998	0.990
Average	0.952	0.877	0.873	0.977	0.936	0.949	0.989	0.961	0.972

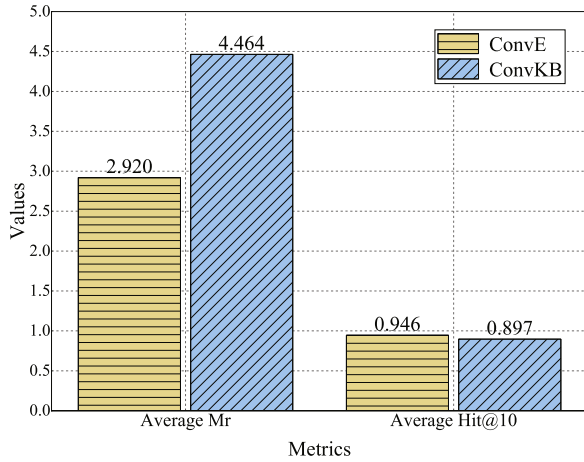


Fig. 11. 2D convolution vs 1D convolution.

Table 10

Predictive scores for IOCs labelling and APT organisation.

Head Entity	Relationship	Tail Entity	Score
0839b3f0a4b28111efc94942436041cb	Associated	APT-C-09	f = 0.928
0839b3f0a4b28111efc94942436041cb	Associated	APT-C-01	f = 0.008
0839b3f0a4b28111efc94942436041cb	Associated	APT-C-06	f = 0.004
0839b3f0a4b28111efc94942436041cb	Associated	APT-C-55	f = 0.006
0839b3f0a4b28111efc94942436041cb	Associated	APT-C-32	f = 0.012

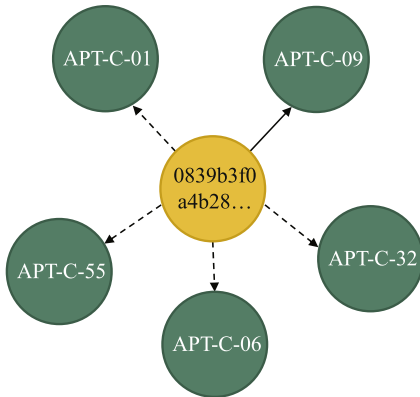


Fig. 12. Visual analysis of IOCs and APT organisation.

APT attacks can be effectively and specifically predicted. We set the threshold to 0.8-if the score of a triple is greater than 0.8, we assume the relationship exists; otherwise, it does not. In this section, using the IOCs tags generated by a particular attack, we predict the associated APT organization, CVE vulnerabilities, CWE weaknesses, and CAPEC attack methods. By calculating the scores of the triples in the test set, the prediction results and their scores are displayed. To demonstrate the feasibility of this method, we perform reasoning on the behaviors of three well-known APT organizations: Molerat (APT-C-08), Lazarus Group (APT-C-26), and White Elephant (APT-C-09). Taking APT-C-09 as an example, we provide a detailed explanation of the reasoning process.

(1)IOCs labelling and APT organisation

From the data in Table 10, it can be observed that only the score of the triplet <0839b3f0a4b28111efc94942436041cb, Associated, APT-C-09> exceeds 0.8. This indicates that there is a relationship between these two entities, leading to the identification of the APT-C-09 APT organization as the perpetrator of this attack. Visual analysis is conducted in Fig. 12, showcasing the predictive results of potentially associated APT organizations.

Table 11

Predictive scores for APT organisations and CVE vulnerabilities.

Head Entity	Relationship	Tail Entity	Score
APT-C-09	Exploits	CVE-2010-3333	f = 0.937
APT-C-09	Exploits	CVE-2017-8759	f = 0.008
APT-C-09	Exploits	CVE-2016-7262	f = 0.003
APT-C-09	Exploits	CVE-2017-12824	f = 0.001
APT-C-09	Exploits	CVE-2015-3652	f = 0.001

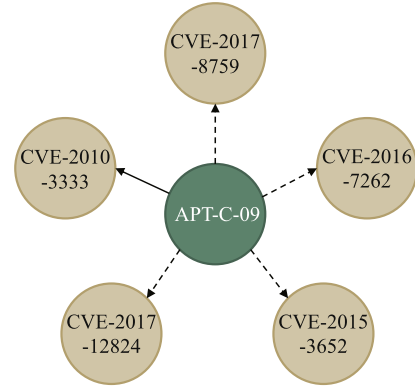


Fig. 13. Visual analysis of APT organization and CVE vulnerability.

Table 12

Predicted scores for CVE vulnerabilities and CWE weaknesses.

Head Entity	Relationship	Tail Entity	Score
CVE-2010-3333	Has	CWE-119	f = 0.934
CVE-2010-3333	Has	CWE-20	f = 0.003
CVE-2010-3333	Has	CWE-22	f = 0.005
CVE-2010-3333	Has	CWE-287	f = 0.002
CVE-2010-3333	Has	CWE-330	f = 0.008

(2)APT organisations and CVE vulnerabilities

From the data in Table 11, it can be observed that only the score of the triplet <APT-C-09, Exploits, CVE-2010-3333> exceeds 0.8. This indicates that there is a relationship between these two entities, leading to the identification that the APT-C-09 attack utilized the CVE-2010-3333 vulnerability. Visual analysis is conducted in Fig. 13, showcasing the predictive results of potentially exploited vulnerabilities.

(3)CVE Vulnerabilities and CWE Weaknesses

From the data in Table 12, it can be observed that only the score of the triplet <CVE-2010-3333, Has, CWE-119> exceeds 0.8. This indicates that there is a relationship between these two entities, leading to the identification that the CVE-2010-3333 vulnerability may have the CWE-119 weakness. Visual analysis is conducted in Fig. 14, showcasing the predictive results of potentially existing vulnerabilities.

(4)CWE Weakness and CAPEC Attack Patterns

From the data in Table 13, it can be observed that the scores of the triplets <CWE-119, Related, CAPEC-45> and <CWE-119, Associated, CAPEC-8> exceed 0.8. This indicates that there is a relationship between the CWE-119 entity and the CAPEC-45 and CAPEC-8 entities. Consequently, it is identified that CWE-119 may be associated with the CAPEC-45 and CAPEC-8 attack patterns. Visual analysis is conducted in Fig. 15, showcasing the predictive results of potentially associated attack patterns.

Based on the analysis of Fig. 16, the effectiveness of the model in APT attack prediction has been verified. Fig. 16(a) presents the predicted attack chain of the model, and a comparison with the real knowledge graph in Fig. 16(b) reveals that the model successfully identified

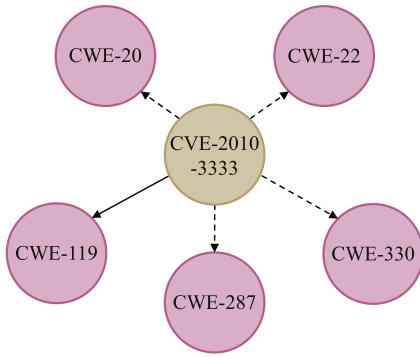


Fig. 14. Visual analysis of CVE vulnerability and CWE weakness.

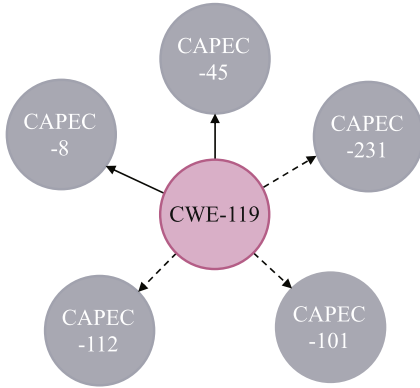


Fig. 15. Visual analysis of CWE weakness and CAPEC attack patterns.

Table 13

Predictive scores for CWE weakness and CAPEC attack patterns.

Head Entity	Relationship	Tail Entity	Score
CWE-119	Related	CAPEC-45	$f = 0.958$
CWE-119	Related	CAPEC-8	$f = 0.963$
CWE-119	Related	CAPEC-101	$f = 0.023$
CWE-119	Related	CAPEC-112	$f = 0.017$
CWE-119	Related	CAPEC-231	$f = 0.008$

APT-C-09 as the attacker. It also accurately predicted the exploited CVE vulnerabilities (such as CVE-2010-3333 and CVE-2017-8759) and the associated CWE weaknesses (such as CWE-119 and CWE-330). Additionally, the model correctly linked potential CAPEC attack patterns used in APT attacks (such as CAPEC-8, CAPEC-231, and CAPEC-45). These results demonstrate that the model not only identifies the attackers and their strategies with precision but also predicts specific vulnerability exploitation and attack methods, showcasing its effectiveness and feasibility in real-world APT attack detection. The high consistency with real attack data highlights the model's significant application value in security defense and threat intelligence analysis.

In this paper, we further predicted the attack behaviors of the two organizations, APT-C-08 (Mandiant's "Malinghua") and Lazarus Group (APT-C-26). As shown in Fig. 17, based on the IOCs left behind, we can infer that the source of the attack originates from APT-C-08 and APT-C-26. By conducting inference on these two organizations, we identified the potential vulnerabilities they might exploit, namely CVE-2017-12824 and CVE-2017-0144. CVE-2017-12824 is a heap-based buffer overflow vulnerability in ImageMagick, while CVE-2017-0144 is a severe vulnerability in the protocol of Microsoft Windows operating systems. Further inference on these vulnerabilities reveals the associated weaknesses, CWE-119 and CWE-20. CWE-119 refers to improper restrictions of operations within the bounds of a memory buffer, which could

result in unintended information exposure due to system misconfigurations. CWE-20 is related to improper input validation, where systems or applications fail to correctly validate user input, leading to potential security vulnerabilities. Building on these identified weaknesses, we can predict the possible attack techniques, both current and future, that may be employed (CAPEC-10, CAPEC-100, CAPEC-261, and CAPEC-250). CAPEC-10 represents buffer overflow attacks, CAPEC-100 refers to command injection attacks, CAPEC-261 is a race condition attack, particularly the "Time-of-Check to Time-of-Use" (TOCTOU) issue, and CAPEC-250 signifies replay attacks. Through this step-by-step inference process, we gain a deeper understanding of the attack strategies used by these organizations, which helps various departments quickly identify existing vulnerabilities and implement effective defenses against such attacks.

6.6. Experimental analysis and discussion

The advantages of the residual multi-head graph attention network (RMultiGAT) and the two-dimensional convolution-based knowledge graph embedding model (ConvE) are evident in both classification and prediction tasks related to APT attacks. Compared with traditional graph neural network models, RMultiGAT benefits from residual connections that preserve critical features in deeper layers and significantly enhance model stability. In addition, the multi-head attention mechanism captures complex dependencies across different dimensions of the attack graph, thereby improving classification accuracy. In contrast to conventional knowledge graph embedding methods, ConvE leverages the local perceptual capacity of two-dimensional convolution to more effectively model the semantic interactions between entities and relations. This yields more compact and interpretable embedding representations, leading to more accurate prediction outcomes. Moreover, GA-ConvE substantially outperforms the direct prediction model (DP-ConvE) in prediction metrics, validating the effectiveness and rationality of the "classification-before-prediction" strategy. The superiority of two-dimensional convolution over one-dimensional convolution in feature extraction and semantic modeling further explains the notable performance improvements observed.

Although GA-ConvE incurs slightly higher runtime and memory consumption compared with certain baseline methods, the overall computational cost remains within an acceptable range. In real-world cybersecurity applications, moderately increasing computational expenditure in exchange for enhanced prediction accuracy and stability constitutes a reasonable and necessary trade-off. Furthermore, the application experiments demonstrate that GA-ConvE can comprehensively reconstruct the APT attack chain—from IOCs to APT organizations, and further to CVE vulnerabilities, CWE weaknesses, and CAPEC attack patterns—highlighting its strong interpretability and practical value for deployment.

7. Challenges

Overall, GA-ConvE not only achieves superior performance in classification and prediction accuracy but also demonstrates notable advantages in terms of interpretability, robustness, and practicality, thereby offering a feasible pathway for intelligent detection of APT attacks and threat intelligence analysis. Nevertheless, this approach still has certain limitations that warrant further investigation in future research.

7.1. Insufficient multi-step reasoning capability

The current model primarily relies on single-step relation prediction; although it performs chained inference reasonably well, attack chains in more complex scenarios may comprise consecutive events spanning multiple stages. A single-step reasoning paradigm struggles to capture such long-range dependencies and lacks support for multi-step knowledge reasoning. Future work should incorporate path-search-based or

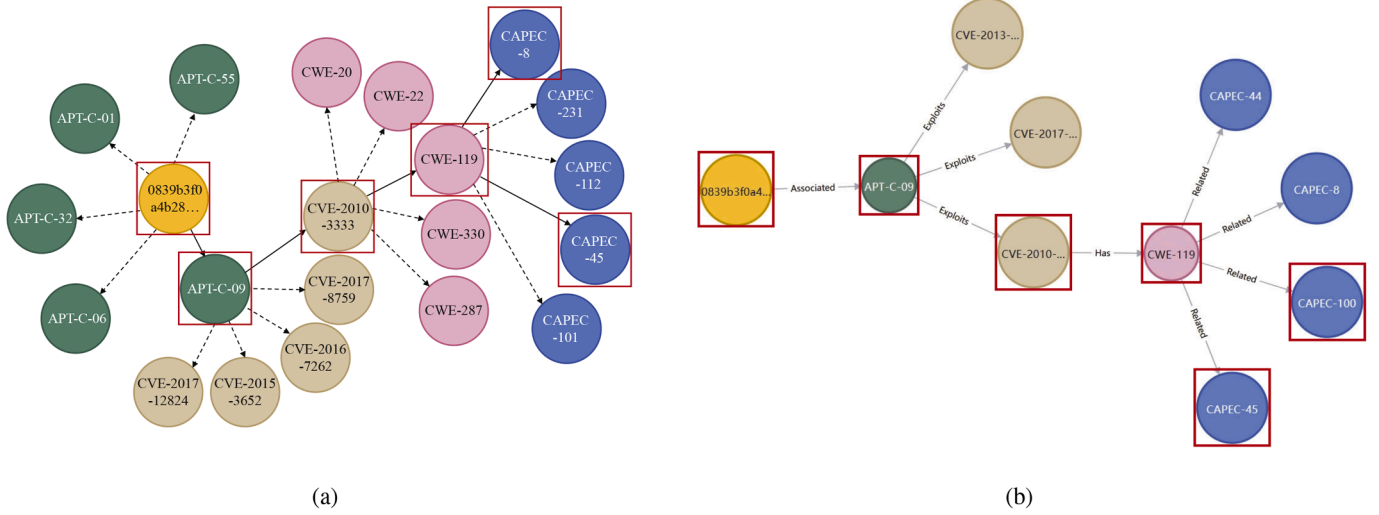


Fig. 16. APT attack visual analysis (a) apt attack predictive visual analysis (b) apt attack knowledge graph (partial).

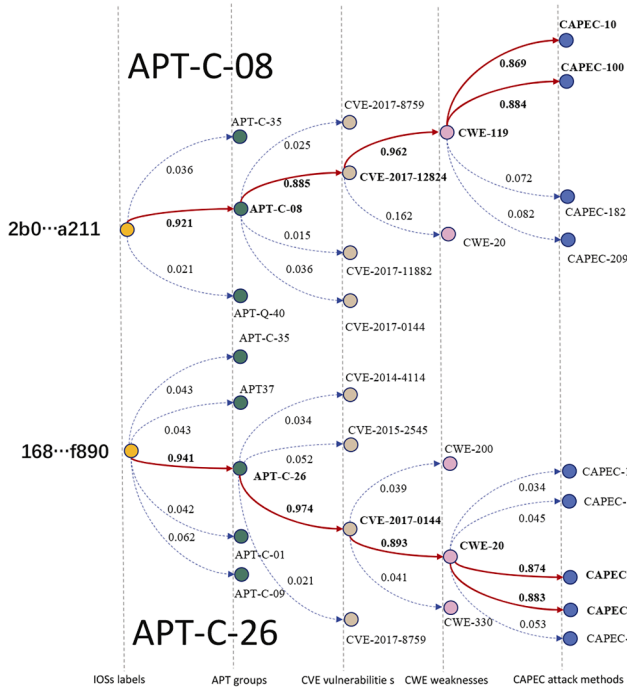


Fig. 17. Prediction results for APT32 and APT-C-01 organizations (partial).

reinforcement learning methods to enhance the model's multi-step reasoning capacity.

7.2. Limited extrapolation and reasoning over unknown knowledge

GA-ConvE relies on a pre-constructed knowledge graph and is unable to directly address novel vulnerabilities, weaknesses, or attack patterns beyond the training data. In real-world network environments, APT attack techniques are continuously evolving, and relying solely on known knowledge may cause the model to fail when confronted with emerging threats. Therefore, integrating knowledge extrapolation with the open-world assumption, and incorporating incremental learning or adversarial learning approaches, will be critical to enhancing the model's practicality.

7.3. Algorithmic complexity and resource consumption

GA-ConvE requires deep graph learning and convolution operations in both classification and prediction stages, resulting in higher overall complexity compared with several lightweight models. Experimental results indicate that its training time and memory consumption are slightly higher than those of the baseline approaches. While this overhead is acceptable in experimental settings, it may pose efficiency bottlenecks in real-time detection or resource-constrained deployment environments (e.g., edge computing devices). Consequently, future research should consider techniques such as model compression, pruning, and knowledge distillation to reduce inference costs.

7.4. Insufficient mining of APT report data

This study primarily constructs the knowledge graph based on APT reports and the MITRE database; however, the issue of insufficient data mining still remains. Certain information contained in APT reports—such as malware characteristics and behavioral features—has not been fully exploited. Relying solely on regular expression matching and entity extraction may lead to low utilization of available information. Future research should incorporate natural language processing (NLP) techniques to extract multi-dimensional features more deeply from unstructured report texts, thereby further enhancing the coverage and accuracy of the knowledge graph.

8. Conclusions

This paper proposes an APT attack prediction model that combines Graph Attention Networks (GAT) and two-dimensional convolution. The model collects recent threat intelligence reports from 12 APT groups, extracting information on various APT attack behaviors, including knowledge-based data (vulnerability databases) and behavior-based data (various IOCs). These data are organized into an APT attack behavior knowledge graph. The RMultiGAT model is used to optimize multi-layer attention networks through residual connections, allowing for the classification of similar APT attacks. At the same time, the GA-ConvE model links APT knowledge and behaviors, performing internal inference training for each APT attack class. This narrows the inference space while enabling more precise extraction of embedded representations for APT local features, achieving more efficient and accurate attack inference. Additionally, since the inference process is built on a knowledge structure, the results are interpretable to a certain degree. Through comparisons between direct prediction and exclud-

ing misclassified results, as well as evaluating the model against other methods, the feasibility and robustness of this approach are validated. Experiments on predicting real attack data demonstrate that the proposed method is more effective in predicting APT attacks in real-world situations.

CRedit authorship contribution statement

Yazhou Du: Validation, Methodology, Formal analysis, Writing - original draft, Writing - review & editing; **Weiwu Ren:** Project administration, Funding acquisition, Validation, Writing - review & editing; **Wenjuan Li:** Data curation, Writing - review & editing; **Minyue Wang:** Data curation, Writing - review & editing; **Wanxiang Wang:** Data curation; **Hewen Zhang:** Data curation; **Mingqi Xia:** Data curation.

Data availability

The data is publicly available at: https://github.com/dyz1209/GA-ConvE_data

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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